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CMPUT 365: Introduction to Reinforcement Learning

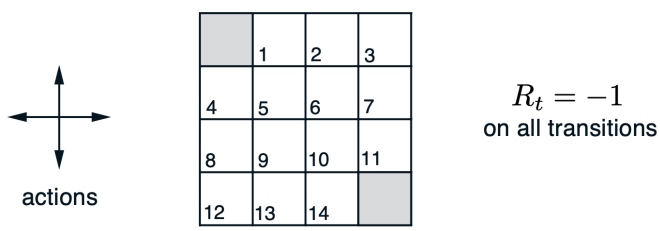
Worksheet 2: Markov Decision Processes (MDPs)

Question 1. Assume you have a bandit problem with 4 actions, where the agent can see rewards from the set $\mathcal{R} = \{-3.0, -0.1, 0, 4.2\}$. Assume you have the probabilities for rewards for each action: $p(r|a)$ for $a \in \{1, 2, 3, 4\}$ and $r \in \{-3.0, -0.1, 0, 4.2\}$. How can you write this problem as an MDP? Remember that an MDP consists of $(\mathcal{S}, \mathcal{A}, \mathcal{R}, P, \gamma)$.

More abstractly, recall that a Bandit problem consists of a given action space $\mathcal{A} = \{1, ..., k\}$ (the k arms) and the distribution over rewards $p(r|a)$ for each action $a \in \mathcal{A}$. Specify an MDP that corresponds to this Bandit problem.

Action space: $\mathcal{A}$.
Set of possible rewards: $\mathcal{R}$.
Set of states: $\mathcal{S} = \{1\}$.
Dynamics model: $p(s', r|s, a) = p(r|a)$ where s' and s always equal 1.
$\gamma = 0$.

Question 2. Consider the 4×4 gridworld shown below.



The nonterminal states are $\mathcal{S} = \{1, 2, \ldots, 14\}$. There are four actions possible in each state, $\mathcal{A} = \{$ `up`, `down`, `right`, `left` $\}$, which deterministically cause the corresponding state transitions, except that actions that would take the agent off the grid in fact leave the state unchanged. In this context, what are the values for the expressions below?

(a) $p(6, -1|5,\texttt{right}) = 1.0$ (b) $p(7, -1|7,\texttt{right}) = 1.0$ (c) $p(10, r|5,\texttt{right}) = 0.0$

Question 3. Prove that the discounted sum of rewards is always finite, if the rewards are bounded: $|R_{t+1}| \leq R_{\max}$ for all t for some finite $R_{\max} > 0$.

$$\left| \sum_{i=0}^{\infty} \gamma^i R_{t+1+i} \right| \qquad \text{for } \gamma \in [0, 1).$$

Hint: Recall that $|a + b| < |a| + |b|$.

$$
\begin{aligned}
\left|\sum_{i=0}^{T-t-1} \gamma^i R_{t+1+i}\right| &\leq \sum_{i=0}^{T-t-1} \left|\gamma^i R_{t+1+i}\right| \\
&= \sum_{i=0}^{T-t-1} \gamma^i \left|R_{t+1+i}\right| \\
&\leq \sum_{i=0}^{T-t-1} \gamma^i R_{\max} \\
&= R_{\max} \sum_{i=0}^{T-t-1} \gamma^i \\
&= R_{\max} \frac{1-\gamma^{T-t}}{1-\gamma} \\
&< \infty
\end{aligned}
$$

When $T \to \infty$:

$$
\begin{aligned}
\lim_{T\to\infty} R_{\max} \frac{1-\gamma^{T-t}}{1-\gamma} &= R_{\max} \frac{1}{1-\gamma} \\
&< \infty.
\end{aligned}
$$

Question 4. Exercise 3.6 from Sutton & Barto (2018).
Dr. Sutton has asked that the solutions to the textbook exercises not be made electronically available to students. You are encouraged to attend office hours, where the teaching assistants can discuss the exercises with you and show you the answers. Alternatively, if you email Dr. Sutton your answers for an entire chapter, he will likely send you the electronic answer sheet for that chapter.

Question 5. Exercise 3.7 from Sutton & Barto (2018).
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Question 6. Exercise 3.8 from Sutton & Barto (2018).
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Question 7. Exercise 3.9 from Sutton & Barto (2018).
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Question 8. Exercise 3.10 from Sutton & Barto (2018).

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